

Hours-of-Service Regulatory Specification for the Rules Engine

This document gives the development team the complete regulatory logic that the Driver Compliance Watch (DCW) rules engine must implement. DCW is a preventive monitoring service: it reads ELD data (read-only) and must identify Hours-of-Service (HOS) violations — and conditions approaching a violation — before they surface in roadside inspections or FMCSA SMS scores. Everything in the engine must be deterministic: the same log data must always produce the same result, with no AI inference in the HOS math.

Four distinct rulesets must be supported, because our client carriers operate under four different regulatory regimes:

- **Ruleset A — Interstate carriers.** Federal HOS rules, 49 CFR Part 395 (§395.3 for property carriers). Applies to any driver whose trip is part of interstate or foreign commerce — including every cross-border movement from Mexico.
- **Ruleset B — Federal 150 air-mile short-haul exception.** 49 CFR §395.1(e)(1) and (e)(2). Interstate drivers who stay within a 150 air-mile radius and meet all conditions are exempt from RODS/ELD and the 30-minute break, but the engine must verify the conditions daily.
- **Ruleset C — Texas intrastate carriers.** 37 Texas Administrative Code §4.12, administered by Texas DPS. Different daily limits (12 driving / 15 on-duty) and a single 70-hour/7-day cycle.
- **Ruleset D — Texas intrastate 150 air-mile exemption.** 37 TAC §4.12(a)(4)–(5). Texas's own short-haul relief with its own conditions (12-hour release, 8 hours off).

Note on "intrastate carriers" generally: intrastate HOS rules are set state by state. Since our client base operates in the Texas/New Mexico border corridor, this specification defines Texas intrastate rules in full. New Mexico adopts the federal HOS rules for intrastate commerce with minor exceptions, so intrastate NM drivers can be run against Ruleset A. If we onboard carriers domiciled in other states, additional state profiles will be specified at that time — the engine should therefore treat the ruleset as a configurable profile assigned per driver, not hard-coded.

Critical design requirement

The applicable ruleset is a property of the DRIVER-DAY, not of the carrier. The same driver can be short-haul exempt on Monday, lose the exemption on Tuesday (drove past 150 air-miles), and run a normal interstate day on Wednesday. The engine must evaluate exemption eligibility every day, per driver, and fall back to the full ruleset the moment a condition fails.

2. Ruleset Selection Logic (Which Rules Apply)

Each driver profile in DCW must carry two configuration flags set by our compliance team at onboarding, plus daily conditions the engine evaluates automatically:

- **operating authority:** INTERSTATE or TX_INTRASTATE. Set by the compliance team based on the carrier's authority and the driver's actual work. Key legal point: "interstate commerce" follows the freight, not truck. A driver who never leaves Texas but hauls freight that originated in Mexico or another state (a continuous movement, e.g., a drayage leg from the Ysleta or BOTA bridge to a Horizon City warehouse) is in interstate commerce and must run Ruleset A. Only freight that both originates and terminates in Texas qualifies for Ruleset C.
- **short haul eligible:** YES/NO. Set by the compliance team — whether the carrier intends to use the 150 air-mile exception for this driver. If YES, the engine evaluates the exemption conditions daily (Rulesets B or D). If any condition fails on a given day, that day is processed under the full ruleset (A or C) and a RODS-required event is raised.

Daily decision flow:

- 1. Determine driver's base regime (A or C) from operating authority.
- 2. If short haul eligible = YES, test that day's data against the exemption conditions (Section 4 or Section 6).
- 3. If all conditions pass → the day is evaluated under the short-haul rules (B or D). If any condition fails → evaluate the full day under the base regime, and flag "exemption lost — RODS required for this day" (federal: §395.1(e)(1) requires a full log for that day).
- 4. Track the 8-days-in-30 counter: a federal short-haul driver who exceeds the exemption conditions more than 8 days in any rolling 30-day period must use an ELD going forward (§395.8(a)(1)(iii)). Raise a distinct alert for this.

3. Ruleset A — Interstate Carriers (49 CFR §395.3, Property-Carrying)

3.1 Who this covers

All drivers of property carrying CMVs in interstate or foreign commerce: GVWR/GCWR or actual weight of 10,001 lbs. or more, or any vehicle placarded for hazardous materials. This is the ruleset for all of our cross-border carriers — every load crossing at BOTA, Ysleta–Zaragoza, Santa Teresa, Tornillo–Guadalupe or Palomas–Columbus is foreign commerce regardless of where the U.S. leg ends.

3.2 The clocks the engine must maintain

Federal HOS has four simultaneous clocks plus a break timer. All must be tracked in parallel; a violation on anyone is a violation.

Rule	When a violation occurs	Detection logic for the engine
11-Hour Driving Limit	Driver accumulates more than 11 hours of Driving status without first obtaining 10 consecutive hours off duty (or a qualifying sleeper-berth equivalent).	Sum all Driving-status minutes since the end of the last valid 10-hour reset. The moment cumulative driving exceeds 11:00, every additional driving minute is in violation. Severity scales with overage (see §8.3).
14-Hour Window	Driver performs ANY driving after the 14th consecutive hour since coming on duty following 10 consecutive hours off. The window runs on the wall clock — off-duty time, lunch, and waiting DO NOT pause it (exception: qualifying sleeper-berth periods, §3.4).	Anchor a window start timestamp at the first on-duty or driving event after each valid 10-hour reset. Any Driving-status segment that begins, continues, or ends after window start + 14:00 is a violation. On-duty (not driving) work after hour 14 is NOT itself a violation — only driving is.
30-Minute Break	Driver drives after accumulating 8 cumulative hours of driving time without at least one 30-minute non-driving interruption. The break can be off duty, sleeper berth, OR on-duty not driving — any consecutive 30 minutes of non-driving status qualifies.	Maintain a driving-since-last-break accumulator. Reset it whenever a consecutive non-driving period of ≥ 30 minutes is found. If the accumulator exceeds 8:00 and Driving status continues, flag from that minute forward.
60/70-Hour Cycle	Drivers drive after accumulating 60 on-duty hours in 7 consecutive days (carriers not operating every day of the week) or 70 hours in 8 consecutive days (carriers operating daily). On-duty = Driving + On-Duty-Not-Driving.	Rolling window: sum ON + D minutes over the trailing 168h (60/7) or 192h (70/8), per the carrier's declared cycle (a per-carrier config value). Driving while the sum $\geq$ the cap is a violation. Off-duty and sleeper time never count toward the sum.
34-Hour Restart	Not a violation source by itself — it is the reset mechanism. 34 or more consecutive hours off duty and/or sleeper berth resets the 60/70 cycle to zero.	Detect any consecutive OFF+SB span $\geq 34:00$; zero the cycle accumulator at the end of that span. There is currently no limit on how often a restart may be used and no 1–5 a.m. requirement (those provisions were removed years ago — do not implement them).

3.3 The 10-hour reset

The 11-hour and 14-hour clocks reset only after 10 consecutive hours off duty, 10 consecutive hours in the sleeper berth, or a combination of consecutive off-duty + sleeper time totaling 10 hours. "Consecutive" means unbroken — a 1-minute on-duty event splits the period and the count starts over. This is the single most important state transition in the engine; log the timestamp of every valid reset.

3.4 Sleeper-berth split provision (§395.1(g))

Drivers may split the required 10 hours off into two qualifying periods instead of one:

- One period of at least 7 consecutive hours in the sleeper berth, AND
- A second period of at least 2 consecutive hours, either off duty, in the sleeper berth, or a combination — the two periods together must total at least 10 hours.

Engine consequences (this is where most ELD vendors get it wrong):

- Neither qualifying period counts against the 14-hour window. After a valid pairing, the 11- and 14-hour calculations restart from the END of the FIRST of the two periods.
- Driving time before and after each qualifying period, added together, must not exceed 11 hours; the recalculated window must not pass hour 14 with driving in it.
- The pairing is evaluated retrospectively — the engine often cannot know whether a 3-hour off-duty block is "period 2 of a split" until the 7-hour berth period appears later. Implement a look-back re-evaluation whenever a qualifying berth period closes.

Note: FMCSA is running pilot programs on additional split flexibility (e.g., a 6/4 split). These are pilots only — do not implement them. A change order will be issued if a final rule ever publishes.

3.5 Exceptions the engine must recognize (they suppress false positives)

Rule	When a violation occurs	Detection logic for the engine
Adverse Driving Conditions (§395.1(b))	Driver may extend the 11-hour driving limit AND the 14-hour window by up to 2 hours when unexpected weather/traffic conditions arise that were not known before dispatch.	Cannot be auto-detected. Implement as a per-day driver/carrier annotation flag. When set, evaluate that day at 13 driving / 16 window, and require a remark on the log. DCW reports must still list these days separately for compliance review — the flag is frequently abused.
16-Hour Short-Haul Exception (§395.1(o))	Once per cycle (i.e., not again until after the next 34-hour restart), a driver who starts and returns to the same work-reporting location for the last 5 duty tours may extend the 14-hour window to 16 hours. The 11-hour driving limit still applies.	Track return-to-same-location for prior 5 tours (needs location data or carrier attestation) and one-use-per-restart. If conditions hold and the flag is used, evaluate the window at 16:00 for that day only.
Personal Conveyance	Off-duty driving for personal purposes while relieved of all work responsibility. Not driving time — does not consume any clock.	Honor the ELD's PC special driving category: PC segments count as OFF. BUT: flag PC segments > 3 hours, PC that moves the truck materially closer to the next load, and PC beginning immediately after the driver exhausts hours — these are the classic falsification patterns and belong in the weekly risk report.
Yard Moves	Driving within a yard/terminal, logged as On-Duty Not Driving.	YM counts as ON (consumes the 14-hour window and the cycle, not the 11-hour driving clock). Flag YM segments at highway speeds (> 32 km/h / 20 mph) or on public roads — falsification indicator.

3.6 Records and "form & manner" violations (§395.8)

Roadside inspections write these up as often as the hour limits, and they hit the same HOS Compliance BASIC. The engine must check daily:

- Missing daily certification — driver has not certified the previous day's record (§395.8: log not current / not certified).
- Missing required fields — carrier name, main office address, shipping document number, trailer number, co-driver, total miles.
- Unassigned driving time — vehicle movement not attached to any driver profile. Every unassigned segment must be annotated or assigned; report all of them weekly.
- Log edits — original vs. edited records, especially edits that convert Driving to OFF/PC/YM after the fact. Read-only access still lets us see the edit history; surface every driving-time edit in the risk report.
- ELD malfunction codes — a driver running on paper past 8 days after a reported malfunction is a violation; track malfunction start dates

4. Ruleset B — Federal 150 Air-Mile Short-Haul Exception (§395.1(e))

This is an exception to the RODS/ELD requirement, not an exemption from hours limits. Short-haul drivers still obey the 11-hour driving limit, the 10-hour reset, and the 60/70 cycle. What changes: no ELD/RODS required (the carrier keeps time records instead), no 30-minute break requirement, and a 14-hour duty-period condition instead of the standard 14-hour window rule.

4.1 CDL drivers — §395.1(e)(1)

ALL of the following conditions must hold for the day. The engine evaluates them daily; one failure = full RODS day under Ruleset A.

Rule	When a violation occurs	Detection logic for the engine
150 air-mile radius	Driver goes beyond a 150 air-mile radius (172.6 statute miles / 277.8 km) of the normal work-reporting location.	Great-circle distance from the work-reporting location coordinates (per-driver config) to every GPS ping. Any ping beyond 150 nautical-based air miles fails the day. Use air miles, not road miles.
Return to reporting location	Driver does not return to the normal work-reporting location and is not released from work within 14 consecutive hours of coming on duty.	Duty period = first on-duty event to final off-duty event of the day. If the span exceeds 14:00, or the final location is not the reporting location, the condition fails.
10 hours off between shifts	Driver starts a new duty tour without at least 10 consecutive hours off duty since the prior tour.	Same reset detection as Ruleset A. Failure = exemption lost for the new day.
11-hour driving limit	Still fully applicable: more than 11 hours driving after the 10-hour reset is a violation, exemption or not.	Identical accumulator to Ruleset A §3.2 row 1.
60/70-hour cycle	Still fully applicable.	Identical rolling-window logic to Ruleset A §3.2 row 4. Time records substitute for logs as the data source when no ELD is present; where the client runs ELDs anyway (our recommendation), use ELD data.

4.2 Non-CDL drivers — §395.1(e)(2)

For short-haul drivers who do not require a CDL (e.g., vehicles 10,001–26,000 lbs GVWR), the radius is also 150 air-miles, with a modified duty window: the driver may not drive after the 14th hour of coming on duty on 5 days of any 7-consecutive-day period, and not after the 16th hour on 2 days of that period. All other conditions (return to location, 10-hour reset, 11-hour driving, 60/70 cycle) apply. The engine needs a per-driver CDL required flag to pick the correct sub-rule.

4.3 The 8-in-30 ELD trigger

High-value preventive alert

§395.8(a)(1)(iii): a driver who does NOT meet the short-haul conditions on more than 8 days in any rolling 30-day period loses the ELD exemption and must run an ELD. Carriers get written up for this constantly and rarely see it coming.

Engine: maintain a rolling 30-day counter of exemption-failure days per driver. Alert at 5 failures (warning), 7 (urgent), 9+ (violation — ELD now required). This alert alone justifies DCW for short-haul fleets.

5. Ruleset C — Texas Intrastate Carriers (37 TAC §4.12)

5.1 Who this covers

Drivers hauling freight that both originates and terminates in Texas, for carriers operating under Texas intrastate authority (TxDMV certificate). Texas DPS adopts the federal safety regulations for intrastate operations but substitutes its own HOS limits. Applicability threshold differs from federal: Texas intrastate rules apply to vehicles with actual gross weight, registered weight, or GVWR over 26,000 lbs; vehicles designed to transport more than 15 passengers; or any vehicle placarded for hazmat. (Farm vehicles have a separate 48,000-lb threshold.) A 12,000-lb box truck running purely intrastate in Texas is outside these HOS rules entirely — but the same truck crossing state or national lines is federal at 10,001 lbs. The engine needs vehicle weight class in the profile.

5.2 Texas intrastate limits and violations

Rule	When a violation occurs	Detection logic for the engine
12-Hour Driving Limit	Driver drives more than 12 hours following 8 consecutive hours off duty.	Same accumulator pattern as federal, with limit 12:00 and reset threshold 8 consecutive hours off (not 10). Enforcement consequence: a driver caught over the 12- or 15-hour limit is placed out of service for 8 consecutive hours — flag these as OOS-risk violations, the highest severity tier.
15-Hour On-Duty Limit	Driver drives after having been ON DUTY 15 hours following 8 consecutive hours off. Unlike the federal 14-hour rule, this is a limit on accumulated on-duty time, not a consecutive wall-clock window — off-duty breaks during the day extend the workday because they do not count toward the 15 hours.	Accumulate ON + D minutes (not OFF, not SB) since the last valid 8-hour reset. Driving after the accumulator reaches 15:00 is a violation. Do NOT reuse the federal window_start wall-clock logic here — this is the single biggest programming difference between the two regimes.
70-Hour / 7-Day Cycle	Driver drives after having worked and/or driven 70 hours in any 7 consecutive days. Texas has one cycle only — there is no 60/7 option.	Rolling 168-hour sum of ON + D. Restart: 34 or more consecutive hours off duty resets the 7-day total, same mechanics as federal.
No 30-minute break rule	Texas intrastate has NO mandatory 30-minute break. Do not raise federal break violations for Ruleset C drivers.	Suppress the break check entirely under this profile.

5.3 Texas sleeper-berth split

Texas allows the 8 consecutive hours off to be accumulated as: (a) a combination of at least 8 consecutive hours of off-duty and sleeper time, or (b) two sleeper-berth rest periods, provided all of the following hold:

- Neither sleeper period is shorter than 2 hours;
- Driving time immediately before and after each rest period, added together, does not exceed 12 hours;
- On-duty time before and after each rest period, added together, includes no driving after the 15th hour;
- The driver cannot return to normal-rule driving without taking at least 8 consecutive hours off, 8 consecutive hours in the berth, or a combination totaling 8.

5.4 ELD note for Texas intrastate

Texas adopted the federal ELD rule for intrastate operations (with phase-in complete years ago), so intrastate carriers over the threshold run ELDs with the device configured to a Texas intrastate ruleset. Most major ELD platforms (Motive, Samsara, Geotab, Omnitracs) expose "Texas Intrastate" as a HOS profile in their APIs — the engine should read the driver's active ruleset from the ELD feed where available and reconcile it against our own profile flag, flagging mismatches (a driver crossing at Zaragoza while his ELD is set to Texas intrastate is itself a finding for the weekly report).

6. Ruleset D — Texas Intrastate 150 Air-Mile Exemption (37 TAC §4.12(a)(4)–(5))

Texas provides its own short-haul relief from RODS for intrastate drivers operating within a 150 air-mile radius of their normal work-reporting location. The carrier maintains driver time records (the "Driver's Time Record" format in the DPS Motor Carrier Guide) instead of logs. Conditions the engine must verify daily:

Rule	When a violation occurs	Detection logic for the engine
150 air-mile radius	Driver operates beyond 150 air-miles of the work-reporting location.	Same great-circle check as Ruleset B.
Return + 12-hour release	Driver fails to return to the work-reporting location OR is not released from work within 12 consecutive hours.	Duty-period span check with a 12:00 limit — note this is tighter than the federal 14-hour release. This is the condition our clients trip most often.
8 consecutive hours off between tours	Less than 8 consecutive hours off duty separating each 12-hour on-duty period.	Reset detection with an 8:00 threshold.
Underlying limits	The 12-hour driving and 70/7 cycle limits continue to apply; exceeding them is a violation regardless of the exemption.	Run the Ruleset C accumulators in parallel; only the RODS requirement is relieved.

If any condition fails on a given day, the driver must produce a full record of duty status for that day and the engine should evaluate the day under Ruleset C in full.

7. Quick-Reference Comparison Table

Parameter	A — Interstate	B — Fed 150-mi	C — TX Intrastate	D — TX 150-mi
Legal citation	49 CFR 395.3	49 CFR 395.1(e)	37 TAC 4.12	37 TAC 4.12(a)(4)-(5)
Driving limit	11 h	11 h	12 h	12 h
Duty limit	14-h consecutive window (wall clock)	14-h duty period, then released (16-h twice/7d if non-CDL)	15 h accumulated on-duty (not wall clock)	12-h consecutive duty period, then released
Off-duty reset	10 h consecutive	10 h consecutive	8 h consecutive	8 h consecutive
30-min break	Yes, after 8 h driving	No	No	No
Weekly cycle	60/7 or 70/8	60/7 or 70/8	70/7 only	70/7 only
Restart	34 h	34 h	34 h	34 h
Sleeper split	7+3 / 8+2 (10 total)	n/a (must return daily)	2+2 min., 8 total	n/a (must return daily)
RODS / ELD	ELD required	Time records; ELD after 8 fails in 30 days	ELD required (TX profile)	Time records
Radius limit	None	150 air-miles	None	150 air-miles
Weight threshold	10,001 lbs GVWR/GCWR or placarded	10,001 lbs	26,001 lbs, >15 pax, or placarded	26,001 lbs, >15 pax, or placarded
OOS penalty noted in rule	OOS until compliant	OOS until compliant	8 consecutive hours OOS for 12/15-h violations	Same as C

8. Cross-Cutting Detection Logic (All Rulesets)

8.1 Time handling

- All HOS math runs in the driver's home terminal time zone (per §395.8's 24-hour period rule), configured per driver. Never mix UTC into limit calculations; convert once at ingestion.
- The "day" for daily records is the carrier's declared 24-hour period starting time (usually midnight, terminal time) — a per-carrier config value.
- ELD duty-status events are minute-granularity; drive time auto-recorded to the nearest minute. Do not round in the carrier's favor — a violation of 1 minute is still reported (severity tiers handle materiality, §8.3).

8.2 Event-driven evaluation

- Re-evaluate all clocks on every new duty-status event, every log edit, and every certification — not on a fixed schedule. Edits can retroactively create or erase violations; the engine must recompute the affected day and all days forward within the cycle window.
- Sleeper-split look-back (§3.4) and exemption-failure fallback (Sections 4 and 6) both require retrospective reprocessing of the current day and, for splits, up to 2 prior days.

- Store both the computed violation and the exact log segments that produced it — the weekly Spanish-language report must show the driver's actual timeline, not just "14-hour violation."

8.3 Severity tiers for reporting

For the weekly risk reports, classify every finding:

- **CRITICAL (OOS-equivalent):** driving beyond any driving/duty limit by more than 15 minutes; false log indicators (PC/YM abuse patterns, driving-time edits); Texas 12/15-hour violations (mandatory 8-hour OOS at roadside); ELD required but not in use.
- **SERIOUS:** violations of 15 minutes or less; missed 30-minute break; cycle limit exceeded; exemption lost without a RODS produced.
- **ADVISORY (preventive):** the core of DCW's value — conditions approaching a limit: driver within 60 minutes of the 11/12-hour or 14/15-hour limit while still driving; cycle above 90%; short-haul failure count at 5+ of 8 in 30 days; recurring near-misses by the same driver (coaching signal).

8.4 Data fields the engine needs per driver profile

- operating authority (INTERSTATE / TX_INTRASTATE); short haul eligible (Y/N); CDL required (Y/N); cycle (60_7 / 70_8 / TX_70_7); home terminal time zone; work reporting location (lat/long); vehicle weight class; hazmat placard flag; ELD ruleset as reported by the ELD platform (for mismatch detection).

9. Assumptions Flagged for Verification

Consistent with our working method, the following assumptions are built into this spec and should be confirmed before the developers freeze the rules engine requirements:

- 1. All pilot clients (Pilot Client 1, Pilot Client 2) are property carriers — passenger-carrier HOS (10-hour driving / 15-hour on-duty federal) is out of scope for v1.
- 2. New Mexico intrastate drivers will be evaluated under Ruleset A (NM adopts federal HOS intrastate). Confirm no NM-specific carve-outs apply to any onboarded client before go-live.
- 3. Oilfield, agricultural, driveaway-towaway, and utility service exceptions (§395.1 various; TX ag exemptions) are excluded from v1. If we sign a carrier using the oilfield waiting-time exception, that is a change order.
- 4. Adverse-driving and 16-hour exception usage will be captured as manual annotations, not auto-detected, in v1.
- 5. Mexico's NOM-087 driver rest rules are out of scope — DCW monitors the U.S. side of the operation only.
- 6. FMCSA sleeper-berth and split-duty pilot programs (active 2026) create no production requirements. Monitor only.